

PRA

Project Risk Analysis



Project: EM600B	Revision: 1.0	Date 22/02/2024
---------------------------	------------------	--------------------

1. Revision list

List of the official revisions validated by static test review meeting

Revisions	Date	Authors	Note
Rev 1.0	22/02/2024	F. Facchin	First release – in progress

2. Distribution list

Roles are mandatory.

PM (Product Manager), RDM (R&D Manager), PL (Project leader), TM (Test manager) have to be present.

Role	Name	Company
PM	Andrea Bernardi	
PL	Marco Meneghel	
RDM	Mauro Perot	
RDM	Fabio Facchin	
OM	Claudio Balcon	
ACQ	Samuele Capraro	
PRD	Luca Reolon	

3. General guidelines

Risk values: lowest (0), highest (5) (The risk value represents the result of the gravity and occurrence consideration)

1. A list of possible mitigation action have to be listed
2. Must be indicated which mitigation actions will be implemented

PRA

Project Risk Analysis



Project: EM600B	Revision: 1.0	Date 22/02/2024
---------------------------	------------------	--------------------

3. The impact of risk has to be described
4. **RV**=Risk Value, **RR**=Risk reduction, **NRV**=New Risk Value

4. R&D – Development

1. Evaluate if all the type tests can be done with the actual tools/instruments
2. Evaluate how much the use conditions are specified or in conflict
3. Evaluate which are the risks (causes) to go in delay with the project delivery
4. Evaluate aspects in which the actual know how is critical
5. Evaluate if all the aspects of the lifetime are estimable
6. Evaluate if the applicable type tests cover all the application aspects
7. Evaluate new HW development circuit (or component)
8. Evaluate new FW development concept
9. Evaluate new MEC development
10. Evaluate certification process
11. Evaluate the risk do not respect the expected investments

Risk	RV	Possible mitigation actions / Notes	RR	Done	NRV
Radiated Emission tests shall be done in external laboratory equipped with anechoic test room. The risk is to discover late problem that require to modify the hw.		Anticipate test using a fw with the minimum implementations			
65A terminal block design according to UL.		Coverage of the terminal requirements already evaluated in EMS project			
Internal temperature of the product: for 65A direct connection versions the value of the shunt can't be too low to respect the accuracy at lower current (from Ist to Imin) and this could impact the maximum rated temperature at Imax.		The design shall minimize the internal dissipation. We are aware about possible derating of the max. continuous current or the limitation of the maximum rated temperature. As soon as a prototype is ready with the minimum functionality environmental test shall be carried on and the limits defined.			

PRA

Project Risk Analysis



Project: EM600B	Revision: 1.0	Date 22/02/2024
---------------------------	------------------	--------------------

Shunt lamella: Availability of limited number of samples for EMS project and long delivery time for first production.	Evaluate the order of further samples for project development. 30 more samples pieces considered in the project investments			
Moulding tools: EM600B moulding tool for top and bottom housings will be managed with the same local supplier of EMS, DCM1 and EM580 project, because the transparent screen is the same and because the supplier is aware about all the critical aspects. The realization of EMS's tools will be in progress at the moment of the EM600 B project approval. The realization of the moulding tools will determine the project timing.	The tool development time is about 5 months for the first trial; particularly attention is needed to the finalization stage: from when we will receive the first trials to when we will approve the parts (max 3 months)			
Design of measurement board: For the measurement part of EM640, same evaluation already done for EMS: - The TI's AFEs AMC131 and ADC131 have been selected to measure currents/voltages. For AV2 version the ability to measure the start current Ist and the accuracy between Ist and Imin are critical and correlated to: - o Amplification, so to the shunt value (can't be too high to avoid thermal issues) - o Oversampling, so to the number of samples required (64 to achieve the 32nd harmonic in the THD calculation)	After preliminary testing we found out a solution to fulfil the accuracy requirements while achieving also the 32 nd harmonic depth for THD measurement. This requires the use of AMC131 version with 3 ADC input.			
Design of communication board: For cost reason, the Espressif ESP32 module has been selected, either for the WiFi and Ethernet connectivity, despite the need of radio communication. It's the first time we use this module, and we don't have specific experience.	In C phase invested time in external consultancy to get some knowledge and evaluate possible critical aspects of the required functionalities.			

PRA

Project Risk Analysis



Project: EM600B	Revision: 1.0	Date 22/02/2024
---------------------------	------------------	--------------------

ANSI C12.1 certification: not possible to provide accurate estimation because of lack of knowledge and still missing information from UL about the applicable tests and cost/timing.	Still pushing UL to get information/quotation for former EM25 project. Evaluate NMI as an alternative but still not accredited for ANSI.			
MID: firmware update functionality	The update of the metrologically relevant FW requires the log of the event, so the use of a clock with battery for back-up. Considering the architecture, also the FW of the communication board is metrologically relevant (driving the LCD) so that in all the MID versions the battery is required to implement the FW update. This has in impact on the cost for those versions without WiFi for which the RTC wasn't expected (battery+RTC's L&M cost is estimated in 1.8€)			
MID: extension T and D requirements: lack of experience considering different protocols to consider (WiFi is considered an open network with stricter requirements) and the firmware update	Some question asked to NMI about the requirements, still waiting for an answer			
MID: risk to go in delay because of long certification time	NMI has been evaluated as notified body for Module B of MID certification			

5. R&D resources

1. Evaluate the skill of the team
2. Evaluate the specific experience of the team

PRA

Project Risk Analysis



Project: EM600B	Revision: 1.0	Date 22/02/2024
---------------------------	------------------	--------------------

- Evaluate the percentage of the resources allocated out of the total available
- Evaluate the dependence on the external development (consultancy)
- Evaluate the reliability of the external development
- Evaluate the capacity to acquire the external know how

Risk	RV	Possible mitigation actions / Notes	RR	Done	NRV
Risk going in delay if engineers leave		In case of emergency situations, it would be needed to consider moving resources from other teams with impact on other committed activity that will be analysed and agreed if needed.			
Lack of authorized engineers to conduct internal UL test due to resignation of PM and the possible retirement of MF in short/medium-term.		New engineers to join the UL team have been identified and the formation process in in progress. Accelerate the formation/accreditation process with the first one.			

6. Purchasing – new components

R&D list the new components, purchasing evaluate the risk value

- Evaluate multiple supplier
- Evaluate component lead time
- Evaluate life cycle
- Evaluate new supplier reliability

Most of the new components are inherited from the EMS project.

Description	Code	Supplier	Supplier's p/n	Distributor
WIFI - BLUETHOOH MODULE	New, See EMS prj	Espressiv	ESP32-MINI-1-N4	EBV

PRA

Project Risk Analysis



Project: EM600B	Revision: 1.0	Date 22/02/2024
---------------------------	------------------	--------------------

INN3072M SWITCHING REGULATOR MINSOP	New, See EMS prj	Power Integration	INN3072M	EBV
C. ELE 47UF 450V 12,5X25 10000H SEE SPEC	2521685, see EMS prj	Nippon Chemicon Aishi	EKXL451ELL470MK30S EHF2WM470W300TG	CSE Sacchi Elettronica
Ethernet RJ45 connector (w/ transformer for ethernet communication)	New, See EMS prj	Juscom Amtek	X_JSM_1390-1117-100LED 5RT1MR50FBN-C8SAP1U-211-R1	Giamper
AMC131M02, isolated $\Sigma\Delta$ ADC, 24bit, 3ch w/ integrated DC/DC (for AV2 version)	New, See EMS prj	TI	AMC131M03DFMR	TI
ADS131M08, $\Sigma\Delta$ ADC 24bit, 8ch (for AV5 version)	New, See EMS prj	TI	ADS131M08IRSNR	TI
Shunt Lamella	New, See EMS prj			
Neutral lamella	New, See EMS prj			
Trasformatore switching	3000416, see EMS prj			

New custom critical components

Description	Code	Supplier	Supplier's p/n	Distributor
Shunt Lamella	New, See EMS prj			
Neutral lamella	New, See EMS prj			
Trasformatore switching	3000416, see EMS prj			
Plastic parts	new			

PRA

Project Risk Analysis


Project:
EM600B

Revision:
1.0

Date
22/02/2024

Already used critical components

Description	Code	Supplier	Supplier's p/n	Distributor
LCD MODULE 128X96 FOR EM580/DCM SEE SPEC	5300040	Yeebo		Rutronik
MICRO STM32G070CBT6 ST SMD LQFP48 SEE SPEC	2625594	STM	STM32G070CBT6TR	Arrow

7. Production – process

1. Evaluate new process concept risk
2. Evaluate the production output capability
3. Evaluate new tools and production apparatus risk
4. Evaluate status of exiting tools and production apparatus
5. Evaluate risk do not respect the target cost
6. Evaluate the impact on the stock
7. Evaluate subcontractors aspects
8. Evaluate duplication of production site aspects
9. Evaluate resources to run the E phase
10. Evaluate needs for operator training

Risk	Risk value	Possible mitigation actions / Notes	RV	Done	NRV
Production tools included in the project approval (2 lines, 1xEM630 + 1xEM640): - 2x laser machine: the same introduced for DCM/EM580 - 2x3-Phase calibrator Calmet C300B c0.05		According to expectations and future re-evaluation of quantities, evaluate the need for additional test equipment to be added (not in this project).			

PRA

Project Risk Analysis



Project: EM600B	Revision: 1.0	Date 22/02/2024
---------------------------	------------------	--------------------

The production capacity of the approved test equipment covers the second year expected quantity.					
Matrix LCD display check (manual or using vision software)		Evaluate pros/cons of vision based on previous production experience			